



COS 221 Practical Assignment 5

- Date Issued: **22 April 2026**
 - Date Due: **26 May 2026** before **11:00 AM**
 - Demo Date: **27 May 2026**
 - Submission Procedure: **Upload to ClickUP**
 - This assignment consists of **11 tasks** for a total of **300 marks** and **30** bonus marks
 - There are **10 marks** assigned to group cohesion for this practical assignment
-

1 Overview

This last practical assignment in COS221 serves to expose you to projects and activities that will be required of you in the second semester of your second year as well as on the third-year level, especially COS301. This project is a group-based project which will require you to use all of your obtained knowledge in the degree so far plus the effort of your group to complete this assignment. This assignment has minimal specifications to allow you to implement all you have learned, but also discover and showcase to the lecturers and yourself what you are capable of achieving as a Computer Science student of the University of Pretoria.

For this assignment, you are welcome to do and implement everything that you think is suitable for this project. This means that you are welcome to do more than what is required, but not less. Where specifications might seem vague to yourself or your group, make the required decisions and assumptions and add them to your uploaded PDF. Simply put, you are required to use your creativity for this assignment and working in groups will make it easier to find ideas and implement a good project. This process is your first introduction to what will be required of you in your final year and industry as a University of Pretoria graduate.

2 Project Scenario

Everybody loves a good vacation, but a great vacation requires extensive planning. From choosing the destination and booking flights to finding accommodation, places to go and what to eat, it can become exhausting. Even though travel agencies can ease the workload, there are still issues around price transparency and the trustworthiness of these agencies. This is where you and your group come in.

Your group has been commissioned to design and develop **Tripistry**, a travel package platform where travellers browse and compare packages and agencies create and curate them. The basic requirements are as follows:

- Travellers should be able to browse different destinations, flights, accommodations, popular tourist attractions and go-to restaurants.
- They should be able to compare prices from different travel agencies, book packages and comment on their experience with the agency.
- Travel agencies should be able to showcase the packages they have available as well as create new packages for travellers.
- Because some people may want to travel in a group but not have enough companions, agencies should allow group trips which they manage and organise.
- The system must enforce distinct roles for travellers and travel agencies, reflected in both the database schema and the application interface. These two user types have different permissions and interactions with the database and **cannot** share the same interface.

The following resources may be helpful in completing this practical. You are welcome to use any other resources applicable to the scenario:

- UI: <https://www.jet2holidays.com/> (Example only, do NOT replicate this UI.)
- Flight datasets: <https://ansperformance.eu/csv/>
- Hotel and accommodation APIs: <https://rapidapi.com/category/Travel>
- Tourist attraction data: <https://opentripmap.io/>
- Restaurant and destination datasets: <https://www.kaggle.com/datasets>

3 Outcomes

After successful completion of this assignment you should be able to:

- Analyse and understand data from multiple sources
- Curate and clean data for use in a relational database
- Design a database schema to be implemented in a RDBMS for the curated data
- Design and build a web-based application and:
 - Execute a connection to an RDBMS from a server-side programming language
 - Query and manipulate a relational database from a server-side programming language
 - Build a web-based interface using HTML, CSS and JavaScript
 - Utilise the web-based interface to query and manipulate a relational database

4 Constraints

1. Your solution must be implemented as a web-based application that interacts with a relational database.
2. You may ask the Teaching Assistants for help but they will not be able to give you the solutions.
3. For the design task and the relational model, you are required to use draw.io (<https://app.diagrams.net/> or <https://draw.io/>) or a similar software to create your design. Unless explicitly stated otherwise, use **Chen's notation** for ER and (E)ER modelling. Failure to adhere to these requirements may result in marks being deducted. This diagram must be included in the submitted PDF.
4. Hand-drawn diagrams will **not** be accepted. Diagrams must be created using appropriate diagramming software.
5. The PDF, database dump, source code, web-based application and git history will be marked.
6. The web-based application:
 - (a) Which runs and performs what it is supposed to will receive full marks
 - (b) Which runs but does not perform as required will receive partial marks
 - (c) Which does not run will be allocated partial marks based on the functionality it would have exhibited
7. You must use a RDBMS alongside the tools and languages required to build a web-based application.
8. You may utilise any text editor or IDE, upon an OS of your choice.
9. The group will be required to demo the project; failure to demo and adhere to the guidelines set out may lead to penalties or a mark of zero.

5 Milestones

To be able to successfully deliver and demonstrate the project on 27 May 2026, you need to adhere to the following milestones:

Date	Marks	Milestone
24-27 April 2026	–	Register your teams. Teams must consist of 5 members .
13 May 2026	10	Tutor Progress Meeting 1: The initial approach should be completed by this meeting tutors will assist with any issues identified in the demonstrated approach. ensure your (E)ER diagrams and relational mappings are completed. Show your database with a few entries of valid data and some queries to your tutor. Note: you are not expected to have the full database finalised at this stage a few sample entries demonstrating progress are sufficient.
20 May 2026	10	Tutor Progress Meeting 2: Demonstrate your initial web-based application that integrates with the database, representing your minimal viable product.
21-22 May 2026	–	Bookings for the team demonstration will open on 21 May 2026 at 11:00 and close on 22 May 2026 at 11:00.
26 May 2026	10	Individual submission due before 11:00 AM: Submit your self-reflection essay (see Task 10).
26 May 2026	260	Group submission due before 11:00 AM: Submit your group project with all stated requirements.
27 May 2026	40	Demonstrate your final project.

Note on Progress Meetings: The two tutor meetings on 13 May and 20 May are each worth **10 marks** and are assessed individually per team member. Attendance of all team members and meaningful participation are required. No marks will be awarded if a team member fails to pitch up for the meetings.

6 Submission Instructions

This assignment has **two separate submissions**:

Group Submission (due 26 May 2026 before 11:00 AM)

You are required to upload a single archive that includes the following files:

- An archive containing your web-based application. If you used a management tool you are required to mention it.
- A PDF containing the answers to the tasks, as well as the link to your GitHub Repository.
- A file or files containing the SQL statements from your database dump to:
 - Create your database,
 - Create the tables in your database,
 - Populate the tables with the data you used.
- Your archive containing your `.git` folder.
- A `readme.txt` or `readme.md` file (which is also used on GitHub) informing the marker what they should do to build and execute your application.

Individual Submission (due 26 May 2026 before 11:00 AM)

Each team member must submit their own **self-reflection essay** (Task 9) as a separate individual upload to ClickUP. This is a one-page essay and must **not** be included in the group archive.

Upload your archive to ClickUP. **No late submissions will be accepted**, so make sure you upload in good time.

7 Online Resources

- Git: <https://git-scm.com>
- Git Helpful Website with commands and more info on Git: <https://www.atlassian.com/git>
- GitHub: <https://github.com>
- PHP: <https://www.php.net>
- W3Schools :<https://www.w3schools.com/>
- Composer: <https://getcomposer.org>
- Platform for developers to learn, share knowledge and build a career: <https://stackoverflow.com/>

8 Assignment Instructions

Task 1: Research (20 marks)

Conduct research on the vacation and travel industry, focusing on online holiday planning and price comparisons. Include a minimum of **four** references (using IEEE or ACM format) and limit your research to one page; the references can continue onto another page. Cover aspects such as existing travel agencies, travel package comparison tools, popular destinations, accommodation providers and tourist attraction platforms.

Task 2: (E)ER Diagram (30 marks)

After you have conducted your research in Task 1, you are now required to construct an (E)ER-diagram of your project and provide the final model in your uploaded PDF. You are required to mention all assumptions you have made during your modelling and any other information you deem necessary. If you made multiple iterations to reach your final (E)ER-diagram, be sure to include all iterations, as well as notes on how each iteration improves on the previous.

Task 3: (E)ER Diagram to Relational Mapping (30 marks)

Apply the steps for converting your (E)ER-diagram into a relational model. Provide a relational mapping of your (E)ER-diagram in your uploaded PDF. Be sure to indicate the conversion for each step, as well as the assumptions you have made and the choice you have made where a conversion could provide multiple solutions.

Task 4: Relational Schema (30 marks)

Design a relational database schema based on your mappings obtained in Task 3 in the form of both a visual diagram and SQL statements targeted towards MariaDB. Ensure that you include, where applicable, all of the following:

- Primary, Secondary and Foreign Keys
- Constraints and Checks
- Data Types and Length Constraints

Task 5: Web-Based Application (55 marks)

Create a web application using PHP, HTML5, CSS and JavaScript to manage and perform the queries you find necessary for **Tripistry**. Your web application must at a minimum be able to:

- Log in and manage two distinct user types: **Travellers** and **Travel Agencies**.
- Allow travellers to browse destinations, flights, accommodations, tourist attractions and restaurants.
- Allow travellers to compare travel packages across different agencies and book a selected package.
- Allow travellers to leave reviews and ratings for agencies and their packages.
- Allow travel agencies to create, edit and delete travel packages and group trip offerings.
- Allow travel agencies to manage the destinations, flights, accommodations, restaurants and attractions associated with a package.
- Sort and filter packages based on various criteria (price, destination, duration, rating, etc.).
- Display a detailed package view including pricing, itinerary, images, agency information and reviews.
- SQL injection prevention.

Task 6: Data (20 marks)

To ensure your web application has sufficient data for testing and demonstration purposes, you are required to populate your database with a substantial amount of travel packages, destinations, flights, accommodations, tourist attractions, restaurants, agencies and user reviews.

You are welcome to augment the given data and/or include all your relevant data in your database by hand, by using a script that generates artificial data, a script that parses data from an API, or by loading data using XML/JSON feeds from relevant websites or APIs (that you find or write yourself).

You must **explain your data population method** and the reasoning behind your choices in your uploaded PDF.

Task 7: Analyse and Optimise (10 marks)

You are required to make use of your chosen RDBMS and tools to analyse at least one of your query execution plans and report on the performance in your uploaded PDF. Thereafter you are required to explain how you would optimise your query, implement your proposed optimisation and report on the performance gains or losses achieved. You are also required to explain why you believe the observed gain or loss in performance was observed.

***Hint:** Ensure that you have sufficient data in your database, or else you might not observe a meaningful difference.*

Task 8: Development (25 marks)

Your project will be assessed not only on the functionality it provides, but also on your overall development practices such as:

- Usage of git (commits, branching strategy, meaningful messages)
- Data validation techniques (client-side and server-side)
- Utilisation of a package manager
- Ease of understanding, spelling and grammar and structure in your code, git commit messages, uploaded PDF and README
- Quality of the overall delivered solution

Task 9: Demo (40 marks)

Your team will be required to demo your solution to the lecturers of COS 221. Booking slots will be opened closer to the time. Note that **all team members are required to be present** for the demo in order to receive marks. In addition, each team member must explain their individual contributions to the project in the uploaded PDF. The demo will be evaluated based on functionality, user experience, presentation quality, and individual contributions.

Task 10: Self-Reflection (10 marks)

This is an **individual submission** and must be submitted separately from the group archive **before 11:00 AM on 26 May 2026** via ClickUP.

Write a one-page reflective essay (approximately 400–500 words) addressing the following:

- Your personal contributions to the group project and how they fit into the overall solution.
- Challenges you encountered during the project and how you overcame them.
- What you learned from working in a group on a large-scale project.
- What you would do differently if you were to start the project again.

Your reflection will be assessed on **depth of insight**, **honesty** and **quality of writing**. This task is worth **10 marks** and is assessed individually, each team member submits their own essay.

Task 11: Bonus Task – Spice It Up (30 marks)

This task is designed to encourage creativity, innovation and independent learning. You may choose one or more of the options below, each of which introduces a new concept or advanced application beyond what was required in the main project. The task and how it was implemented must be documented and explained in your submission. You may also do something not listed below, it will be up to the discretion of the markers to determine if your chosen innovation counts as something impressive.

1. Enhanced UI/UX

Create an **interactive** traveller-side dashboard showing the following:

- Top-rated packages
- Agency statistics
- Review data from frequent travellers

You may also consider features such as skeleton loading screens and micro-animations for a polished feel.

2. Security Features

Conduct a basic security audit and implement two or more improvements, such as:

- Secure password hashing (elaborate on your choice of hashing algorithm and techniques)
- Login attempt rate-limiting
- CSRF/XSS protection
- Monitoring, through audit or error logs
- Session management improvements

3. Algorithmic Thinking

Because not every traveller has a group, design an algorithm that matches solo or small-party travellers into compatible groups for agency group trips. Your matching criteria should be informed by data already present in your schema (e.g. destination preference, budget, travel dates) and the algorithm's logic, complexity and trade-offs should be discussed in your PDF.

4. AI Integration

Meaningfully integrate AI into your application. Simply connecting to a model's API will earn you no marks in this task. Consider the following:

- Implementing sentiment analysis for reviews
- Trip summaries or descriptions from structured package data

5. Advanced Feature

Design and implement one unique feature not already required. Examples:

- A recommendation engine that suggests travel packages based on user history or preferences
- An interactive map view of destinations and attractions
- An option to integrate trip plans with a traveler's calendar

6. Advanced Git Features

Version control and automation are essential practices in modern software development. Implement Continuous Integration (CI) using GitHub Actions to streamline your development workflow and ensure code quality through automated testing.

Note: This task will require a proper git branching strategy to work well.

IMPORTANT NOTES:

1. Please refer to the rubric on the last page for the detailed allocation of marks.
2. You will be required to demo your project.
3. If you do not demo, you will receive 0.
4. Attendance at **both** tutor progress meetings (13 May and 20 May) is required to receive the associated marks.
5. The **self-reflection essay** (Task 10) is an **individual submission** and must be submitted separately before 11:00 AM on 26 May 2026.

Research	20
General overview and explanation	4
Types or categories of travel products and agencies explained	4
Information on how content is rated or categorised	4
Additional features/information (Recommendations, Group Trips, User Reviews)	4
References	4
(E)ER Diagram	30
Entities and Attributes	10
Complex and Derived Attributes	10
Relationships and Cardinality	10
Mapping	30
Regular Entity Types	3
Weak Entity Types	3
1:1 Relationships	3
1:N Relationships	3
M:N Relationships	3
Multivalued Attributes	3
N-ary Relationships	3
Specialisation and Generalisation	3
Unions	3
Correctness	3
Relational Schema	30
Visual Representation	10
Primary, Secondary and Foreign Keys	5
Constraints and Checks	5
Data Types and Length Constraints	5
Correctness	5
Web-Based Application	55
Functional Requirements (traveller & agency features)	15
Package and Booking Management Features	10
User Account and Profile Management (two distinct interfaces)	10
Sorting and Filtering Options	10
Queries	10
SQL Injection Prevention	5
Sample Data	20
Explanation of Data Sources	5
Data Entry Methods (Script or Manual)	5
Quality and Relevance of Data	10
Analysis and Optimisation	10
Explanation	2
Optimisation	3
Interpretation	5
Development	25
Version Control (Git)	5
README Documentation	5
Overall Quality and Impression	15
Demo	40
Functionality and User Experience	20
Presentation Quality	20
Group Cohesion	10
Progress Meetings	20
Tutor Meeting 1 13 May 2026 (database progress & initial approach)	10
Tutor Meeting 2 20 May 2026 (minimal viable product demo)	10
Self-Reflection	10
Depth of insight and personal reflection	4
Identification of challenges and lessons learned	3
Quality of writing and structure	3
Total	300
Bonus Task	30